

Assignment 3

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Question 1

(a) The **Retrograde motion**.

Reason: When the planet moves on the epicycle, the position of the planet on the constellation will appear to move backward if the speed of the planet on the epicycle is greater than that of the center of the deferent.

(b) To explain **the non-uniform apparent speed** of the planets.

Reason: Ptolemy believes that the speed of the planets should be uniform. The eccentric adjust the center of the orbit, so that the planet moves at a uniform speed, while the equant adjust the center of motion for the planet to move at a constant angular speed. By combining the eccentric and the equant, Ptolemy was able to fix the observation into the model.

Question 2

(a) The elongation is 30°

Reason: The angle formed by the planet, the earth, and the sun takes up to $1/12$ of a circle, so the elongation is $360^\circ/12 = 30^\circ$.

(b) Yes, the period T_P is **one sidereal year**.

Reason: This is because for superior planets, the line from the earth to the sun is always parallel to the line from the center of the epicycle and the planet to fix the observation of the time of the retrograde motion. Therefore, T_P is equal to the period of the sun, which is a sidereal year.

(c) Yes, the period T_O is **one sidereal year**.

Reason: This is because for inferior planets, the sun, the earth, and the center of the epicycle must be on a straight line to fit the observation of the limited elongation. Therefore, T_O is equal to the period of the sun, which is a sidereal year.

(d) The ratio is **1/2**.

Reason: Suppose the maximum elongation is θ , then we have $\sin \theta = \frac{R_{ep}}{R_{def}}$. From the figure given, we can observe that the maximum elongation is 30° . Since $\sin 30^\circ = 1/2$, the ratio is $1/2$.

Question 3

(a) In Ptolemy's system, the order is supported by **philosophy**.

Reason: If the size of the deferent changes, but the ratio of the radius of the deferent and the epicycle, and the period of the deferent and the epicycle do not change, the tropical period is the do not change. Therefore, Ptolemy's order is based on philosophy.

(b) In Copernicus's system, the order is supported by **observation**.

Reason: The distance can be measured by the tropical period mathematically. He ordered the planets based on the distance calculated by the observation.

Question 4

(a) He **does not have evidence directly**, but his hypothesis lead to a **more elegant explanation** of the motion of the planets.

Reason: His model is much simpler than Ptolemy's model since he abandoned the epicycle, eccentric, and equant. However, this cannot be direct evidence for the moving earth hypothesis.

(b) **No**.

Reason: In Copernicus's era, there is no evidence supporting the moving earth hypothesis, which is against the common belief. Besides, for accuracy, his model is mathematically equivalent to Ptolemy's model. Therefore, we cannot say that his model is more scientific.